

## Task 1 – SSL/TLS in Paytm or Zomato

When using applications such as Paytm or Zomato over HTTPS, TLS establishes an encrypted connection between the user's device and the application's server. The server presents a digital certificate that helps the browser verify the server's identity. After the TLS handshake, encryption keys are established and sensitive information such as login details, payment information, and personal data is encrypted during transmission. The padlock icon indicates that the browser has established a secure HTTPS connection and that the certificate passed the browser's trust checks.

## Task 2 – Generate a Self-Signed SSL Certificate

Step 1: Install Apache and OpenSSL

```
sudo apt update
```

```
sudo apt install apache2 openssl -y
```

Check Apache:

```
sudo systemctl status apache2
```

If necessary:

```
sudo systemctl start apache2
```

Step 2: Create a directory for the certificate

- ```
sudo mkdir -p /etc/apache2/ssl
```

Step 3: Generate a private key and self-signed certificate

- ```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
```
- ```
-keyout /etc/apache2/ssl/localhost.key \
```
- ```
-out /etc/apache2/ssl/localhost.crt
```

During the questions, use:

- Common Name: localhost

The important files are:

- localhost.key → Private key
- localhost.crt → SSL/TLS certificate

Step 4: Enable Apache SSL

- `sudo a2enmod ssl`
- `sudo a2ensite default-ssl`

Edit the SSL configuration:

- `sudo nano /etc/apache2/sites-available/default-ssl.conf`

Make sure it contains:

SSLEngine on

SSLCertificateFile /etc/apache2/ssl/localhost.crt

SSLCertificateKeyFile /etc/apache2/ssl/localhost.key

Step 5: Restart Apache

- `sudo systemctl restart apache2`

Step 6: Test the website

Open:

- <https://localhost>

Task 3 – Let's Encrypt and Certbot

Important Note

There is a technical limitation with this task:

- Let's Encrypt does not normally issue certificates for localhost.

A publicly trusted Let's Encrypt certificate requires control over a publicly registered domain name. Therefore, you cannot simply run Certbot for:

- `https://localhost`

and receive a publicly trusted Let's Encrypt certificate.

For a real Let's Encrypt demonstration, use a domain you control, for example:

- example.yourdomain.com
- pointing to your server.

Install Certbot

For Apache:

- sudo apt update
- sudo apt install certbot python3-certbot-apache -y

Then:

- sudo certbot --apache -d example.yourdomain.com\

Certbot can automatically configure Apache for HTTPS.

Test:

- <https://example.yourdomain.com>

Task 4 – Certificate Information

Open:

- https://localhost

Click the browser's certificate/security information.

Record the following three pieces of information:

Task 5 – Troubleshooting SSL/TLS Error

Error: NET::ERR\_CERT\_AUTHORITY\_INVALID  
Problem

The browser displayed:

☐ NET::ERR\_CERT\_AUTHORITY\_INVALID

when accessing:

- https://localhost

Cause

The certificate was self-signed and was not issued by a Certificate Authority trusted by the browser.

### Troubleshooting Steps

1. Checked whether Apache was running:
  - `sudo systemctl status apache2`
2. Checked that SSL was enabled:
  - `sudo a2enmod ssl`
3. Checked the Apache SSL configuration:
  - `sudo nano /etc/apache2/sites-available/default-ssl.conf`
4. Verified that the certificate paths were correct:
  - `SSLCertificateFile /etc/apache2/ssl/localhost.crt`
  - `SSLCertificateKeyFile /etc/apache2/ssl/localhost.key`
5. Restarted Apache:
  - `sudo systemctl restart apache2`
6. Opened:
  - <https://localhost>